

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) CO2 Emissi...Canada.csv

CO2 Emissions_Canada.csv(text/csv) - 476091 bytes, last modified: 2/27/2026 - 100% done
Saving CO2 Emissions_Canada.csv to CO2 Emissions_Canada (1).csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
# Load dataset
df = pd.read_csv("CO2 Emissions_Canada.csv")

# Display first 5 rows
df.head()
```

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Fuel Consumption Comb (mpg)	Emis
0	ACURA	ILX	COMPACT	2.0	4	AS5	Z	9.9	6.7	8.5	33	
1	ACURA	ILX	COMPACT	2.4	4	M6	Z	11.2	7.7	9.6	29	
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	6.0	5.8	5.9	48	
3	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6	Z	12.7	9.1	11.1	25	
4	ACURA	RDx AWD	SUV - SMALL	3.5	6	AS6	Z	12.1	8.7	10.6	27	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
# Feature and Target
X = df[['Engine Size(L)']] # Feature
y = df['CO2 Emissions(g/km)'] # Target
```

```
# Fit Linear Regression
model = LinearRegression()
model.fit(X, y)

# Extract coefficient ( $\beta_1$ )
beta_1 = model.coef_[0]

print("Original Regression Coefficient ( $\beta_1$ ):", beta_1)
```

```
Original Regression Coefficient ( $\beta_1$ ): 36.77731518641943
```

```
B = 2000
n = len(df)

bootstrap_betas = []

for i in range(B):
    # Resample with replacement
    sample = df.sample(n, replace=True)

    X_sample = sample[['Engine Size(L)']]
    y_sample = sample['CO2 Emissions(g/km)']

    model = LinearRegression()
    model.fit(X_sample, y_sample)

    bootstrap_betas.append(model.coef_[0])

bootstrap_betas = np.array(bootstrap_betas)
```

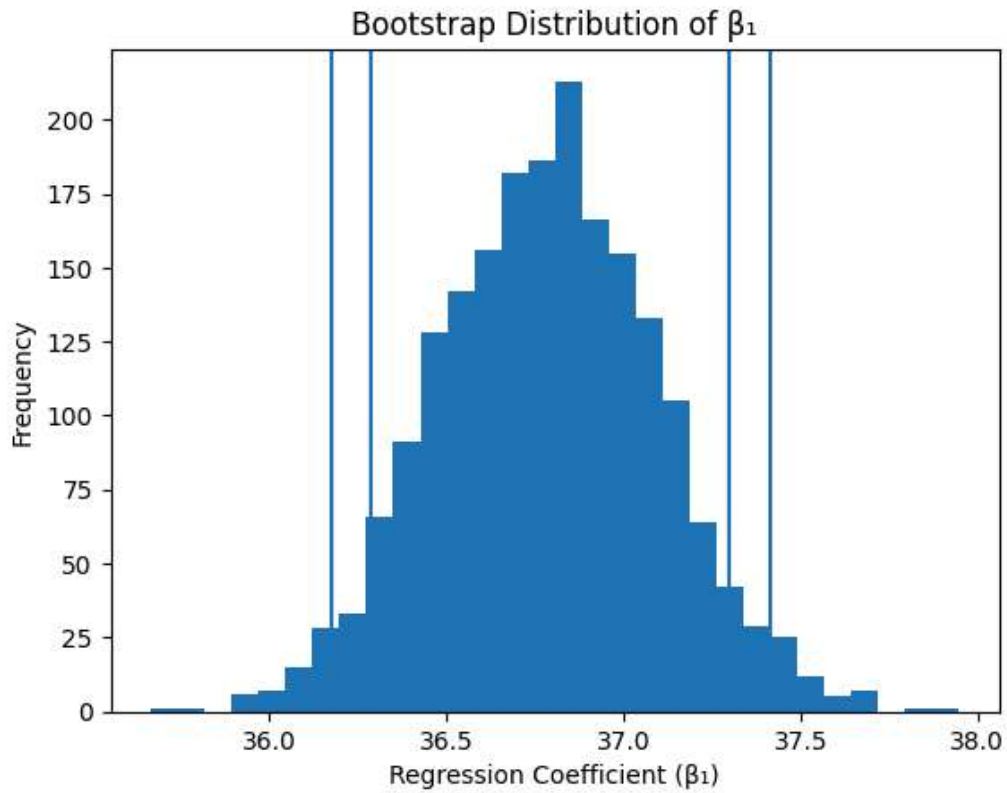
```
# 90% Confidence Interval
ci_90 = np.percentile(bootstrap_betas, [5, 95])

# 95% Confidence Interval
ci_95 = np.percentile(bootstrap_betas, [2.5, 97.5])
```

```
print("90% Confidence Interval for  $\beta_1$ :", ci_90)  
print("95% Confidence Interval for  $\beta_1$ :", ci_95)
```

```
90% Confidence Interval for  $\beta_1$ : [36.28634704 37.29681204]  
95% Confidence Interval for  $\beta_1$ : [36.17816778 37.41577767]
```

```
plt.hist(bootstrap_betas, bins=30)  
plt.axvline(ci_90[0])  
plt.axvline(ci_90[1])  
plt.axvline(ci_95[0])  
plt.axvline(ci_95[1])  
plt.title("Bootstrap Distribution of  $\beta_1$ ")  
plt.xlabel("Regression Coefficient ( $\beta_1$ )")  
plt.ylabel("Frequency")  
plt.show()
```



The bootstrap 90% and 95% confidence intervals for the regression coefficient β_1 do not include zero. Therefore, engine size significantly affects CO₂ emissions. The 95% confidence interval is wider than the 90% interval, as expected